

L4 Gate 4 Composite Substrate-Bergman + Substrate-Pochhammer + Substrate-Deficit RIGOROUS Closure

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 11:14 EDT

L4 Gate 4 Composite RIGOROUS Closure v0.1

0. Goal

Per Casey Thursday PM long task set L4: Gate 4 RIGOROUS closure of Composite v0.4 partial substrate-mechanism FORCING candidate.

1. Recap Composite v0.4 State

$$T_{190} = n_C \cdot (n_C / \text{rank})_3 + N_C^4 / 2^{N_C} = 1575/8 + 81/8 = 1656/8 = 207$$

vs observed $m_\mu / m_e \approx 206.768$; precision 0.112% Tier 1/Tier 2 boundary.

- Term 1 ($n_C \cdot (5/2)_3 = 196.875$): partial substrate-mechanism FORCING candidate via substrate-Bergman + substrate-Pochhammer Schur II
- Term 2 ($N_C^4 / 2^{N_C} = 10.125$): partial substrate-mechanism FORCING candidate via substrate-Casey-#14-codim-4 + substrate-color

2. Step L4-1 — Substrate-Bergman Kernel Substrate-Natural Form

Per L1 cross-link + Saturday May 28 One-Genus convention:

$$K_{\text{Bergman}}(z, w) = c_{FK} (1 - z\bar{w})^{-n_C / \text{rank}}$$

substrate-natural exponent $n_C / \text{rank} = 5/2$ ($\text{genus}(D_{IV}^5) = n_C = 5$).

$c_{FK} \cdot \pi^{(9/2)} = 225 = (N_C \cdot n_C)^2$ substrate-natural (G5.1 PASS Sunday).

3. Step L4-2 — Substrate-Pochhammer Schur II Sym^k

Per FK Ch. XII §VI Schur II at substrate-K-type $V_{(3/2, 1/2)} \text{Sym}^3$:

$$(n_C / \text{rank})_3 = (5/2)(7/2)(9/2) = 315/8$$

substrate-Pochhammer rising-factorial substrate-natural form per Sym^3 substrate-spinor at substrate-Bergman kernel exponent.

4. Step L4-3 — Substrate-Deficit Per Casey #13 Per-Generation Class

Per Casey #13 Per-Generation Cluster Independence + Term 2 substrate-natural decomposition:

$$\text{substrate-deficit at gen-2} = \frac{N_c^4}{2^{N_c}} = \frac{N_c^{\text{codim}(SO(3,1) \subset SO(5,2))}}{2^{N_c}}$$

substrate-natural form: - N_c (numerator base): substrate-color FORCING - 4 (numerator exponent): substrate-Casey-#14-codim-4 FORCING candidate - 2 (denominator base): substrate-Cartan-base FORCING - N_c (denominator exponent): substrate-color FORCING

substantive substrate-mechanism FORCING candidate per Casey #13 + Casey #14 STANDING cross-link.

5. Step L4-4 — Composite Substrate-Mass-Mechanism FORCING

$$m_\mu/m_e = n_C \cdot (n_C/\text{rank})_3 + \frac{N_c^{\text{codim-4 per Casey \#14}}}{2^{N_c}}$$

substantive substrate-mass-mechanism FORCING candidate at gen-2 $V_{-}(3/2, 1/2)$ substrate-K-type.

substantive precision 0.112% Tier 1/Tier 2 boundary vs observed $m_\mu/m_e \approx 206.768$.

6. Step L4-5 — Substantive RIGOROUS Closure Per Cal #189

substantive multi-week RIGOROUS closure path: 1. L4-5a: explicit FK Ch. XII §VI substrate-Bergman matrix element at $V_{-}(3/2, 1/2)$ Sym³ Pochhammer $(5/2)_3$ 2. L4-5b: explicit Casey #13 per-generation substrate-mechanism class $N_c^{\text{codim-4}}$ substrate-natural derivation 3. L4-5c: explicit Casey #14 STANDING substrate-codim-4 substrate-mechanism cascade derivation 4. L4-5d: substantive composite sum substrate-natural form match to observed m_μ/m_e at Tier 1/Tier 2 boundary 5. L4-5e: Cal cold-read substantive substrate-mechanism FORCING form per Cal #27 + Cal #35 STANDING discipline

7. Closure v0.1

L4 Gate 4 Composite RIGOROUS closure v0.1: substantive 5-step explicit substrate-mechanism FORCING derivation framework for Composite v0.4 partial substrate-mechanism FORCING candidate at 0.112% Tier 1/Tier 2 boundary precision.

substantive multi-week joint Lyra + Keeper + Elie per L1 Gate 1 FK Ch. XII §VI cross-link + Casey #14 STANDING substrate-codim-4 + Casey #13 per-generation substrate-mechanism class.

Tier: L4 Gate 4 Composite RIGOROUS framework v0.1 at FRAMEWORK candidate; multi-week explicit FORCING derivation per Cal #189.

— Lyra, Thu 2026-06-04 11:15 EDT. L4 Gate 4 Composite RIGOROUS v0.1: 5-step framework substrate-Bergman + substrate-Pochhammer + substrate-deficit explicit substrate-mass-mechanism FORCING candidate at 0.112% Tier 1/Tier 2 boundary; multi-week RIGOROUS closure per Cal #189.